Marketplace analytics: what drives repeat purchases and 5-star reviews

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Five questions a marketplace analyst answers every week, worked end to end on 91,484 real orders with SQL (DuckDB), Python (tests, regression, RFM, forecast) and Power BI. Nothing is simulated.

REAL · PAGE 1

+161%

GMV growth, Jan-Jul 2018 vs 2017. Yet 97% of customers bought only once.

REAL · PAGE 2

2.2 vs 4.3

stars for late vs on-time deliveries. Late orders are 7% of orders but 33% of 1-2 star reviews.

REAL · PAGE 3

18%

of sellers make 80% of GMV. 32 big sellers with poor ratings are the ones to fix first.

REAL · PAGE 5

~25,400

orders forecast for Q4 2018 (+46% on Q4 2017), with a 22% weekly hold-out error.

Business questions

1 · Growth How does the marketplace grow, and do customers come back?
2 · Delivery Do late deliveries destroy ratings and repurchase?
3 · Sellers Which sellers carry the marketplace, and which put it at risk?
4 · Customers Who buys what, and how should customers be segmented?
5 · Forecast How many orders next quarter, and what should be done first?

Recommendation: cut lateness first (halving it removes ~14% of all 1-2 star reviews), starting with the 32 big, poorly rated sellers; run a second-order campaign for one-time buyers; acquire through the categories that bring customers back.

Data & methods

REAL Olist public dataset, Brazil: 99,441 orders, 96,096 customers, 3,095 sellers, 9 related tables.
CLEANING customer_unique_id as the customer key; cancelled orders out of GMV; window Jan 2017 - Jul 2018 (thin start, data tapers off after 21 Aug 2018).
SQL monthly growth, censoring-safe cohorts, delay vs review, seller scorecard, RFM (DuckDB).
Python Mann-Whitney + bootstrap, logistic regression, Wilson intervals, forecast checked on a 12-week hold-out.
Power BI star schema (4 facts, 5 dimensions), 49 DAX measures, one question per page.

How to read this report

Titles state the conclusion of the page. **GMV** item price + freight, in Brazilian reais (BRL). **Blue** = data, **grey** = comparison, **red** = late or below threshold; dashed lines mark thresholds. **Code** SQL, notebook and Power BI project are in the repository.

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GMV grew 161% in a year, but only 3.0% of customers ever ordered twice

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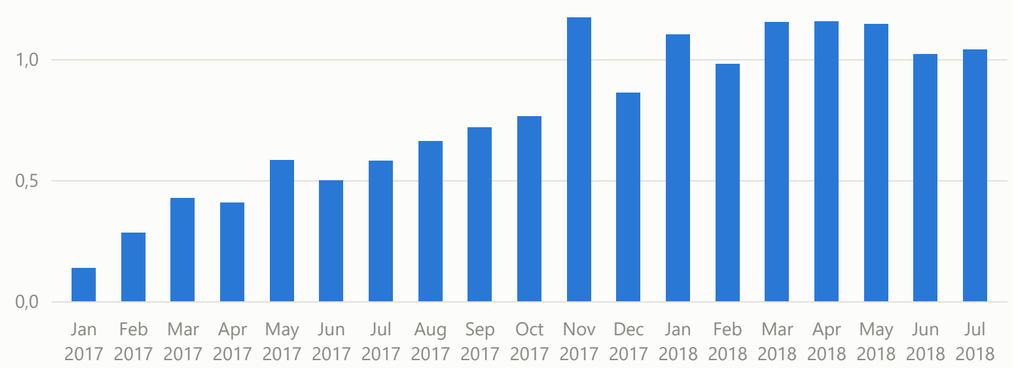
97% of orders are a customer's first order, and only 0.5% of a cohort orders again the following month: growth comes from acquisition, not retention. A cohort is only measured on months it has fully reached.

GMV, BRL million	Valid orders	Customers (customer_unique_id)	Customers with 2+ orders
14,7	91 484	88 494	3,0%

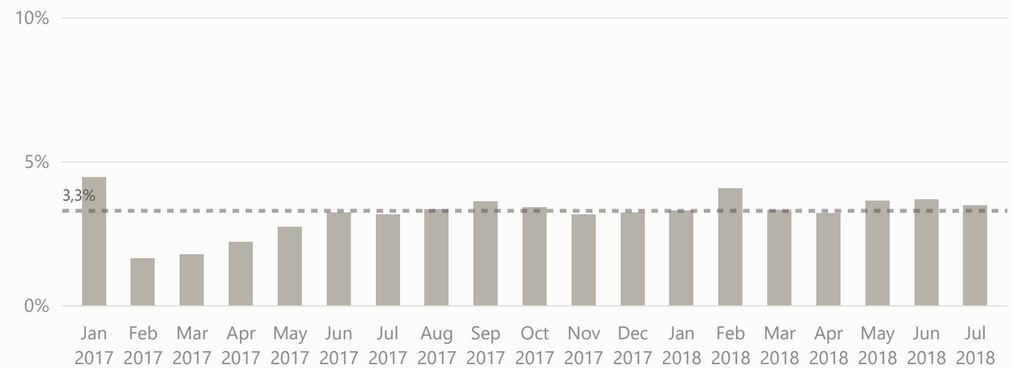
Cohort retention: share of each monthly cohort ordering again N months after its first order (blank = not reached yet)

First order	1	2	3	4	5	6	7	8	9	10	11	12
Jan 2017	0,40%	0,27%	0,13%	0,40%	0,13%	0,40%	0,13%	0,13%	0,00%	0,40%	0,13%	0,66%
Feb 2017	0,24%	0,30%	0,12%	0,41%	0,12%	0,24%	0,18%	0,12%	0,18%	0,12%	0,30%	0,12%
Mar 2017	0,51%	0,35%	0,39%	0,35%	0,16%	0,16%	0,31%	0,35%	0,08%	0,39%	0,12%	0,23%
Apr 2017	0,60%	0,22%	0,17%	0,30%	0,26%	0,34%	0,30%	0,30%	0,17%	0,26%	0,09%	0,09%
May 2017	0,48%	0,48%	0,40%	0,31%	0,34%	0,42%	0,14%	0,25%	0,25%	0,25%	0,34%	0,25%
Jun 2017	0,45%	0,35%	0,39%	0,26%	0,39%	0,35%	0,23%	0,13%	0,23%	0,32%	0,32%	0,16%
Jul 2017	0,52%	0,34%	0,24%	0,29%	0,21%	0,31%	0,10%	0,18%	0,26%	0,21%	0,29%	0,13%
Aug 2017	0,68%	0,34%	0,27%	0,36%	0,53%	0,29%	0,27%	0,15%	0,15%	0,24%	0,19%	
Sep 2017	0,69%	0,54%	0,29%	0,44%	0,22%	0,22%	0,25%	0,29%	0,17%	0,27%		
Oct 2017	0,71%	0,25%	0,09%	0,23%	0,20%	0,20%	0,36%	0,27%	0,20%			
Nov 2017	0,56%	0,39%	0,17%	0,19%	0,18%	0,11%	0,19%	0,13%				
Dec 2017	0,22%	0,28%	0,35%	0,26%	0,20%	0,17%	0,02%					
Jan 2018	0,33%	0,36%	0,29%	0,29%	0,16%	0,17%						
Feb 2018	0,36%	0,39%	0,30%	0,27%	0,22%							
Mar 2018	0,40%	0,30%	0,29%	0,13%								
Apr 2018	0,58%	0,31%	0,24%									
May 2018	0,52%	0,26%										
Jun 2018	0,42%											

GMV by month, BRL million



Share of orders placed by returning customers (dashed = 3.3% overall)



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A late delivery costs 2 stars: 7% of orders arrive late but cause 33% of all 1-2 star reviews

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Average review 4.3 on time vs 2.2 late (Mann-Whitney p < 0.001; bootstrap 95% interval of the gap 2.02-2.10 stars). The effect on repurchase is not detectable (odds ratio 0.84, p = 0.24): correlation, not proof of cause.

Delivered after the promised date

6,8%

Average review, on time

4,3

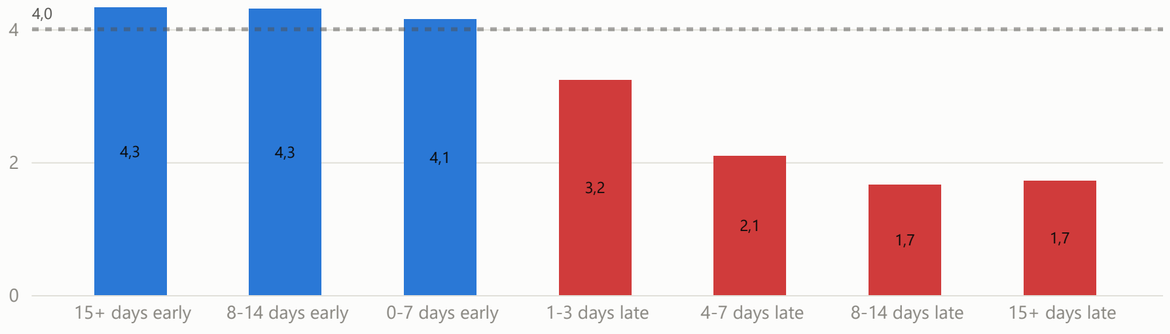
Average review, late

2,2

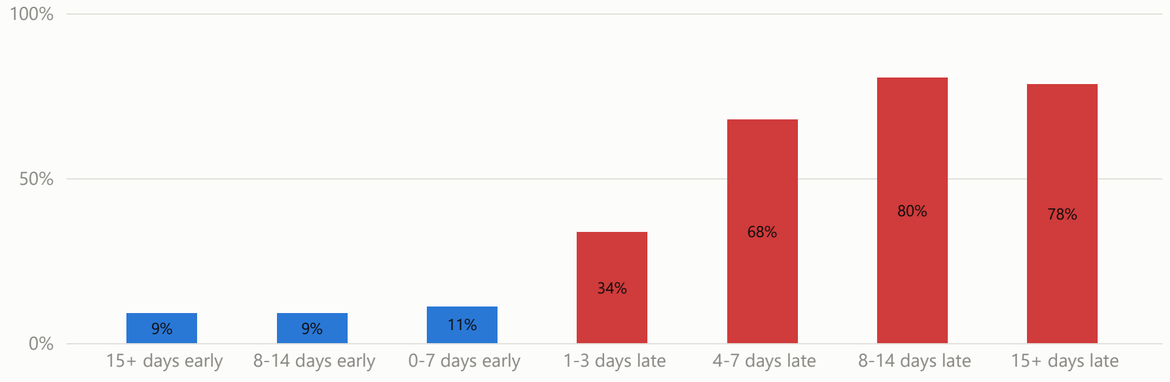
1-2 star reviews among late orders

64%

Average review by delivery delay vs promised date (dashed = 4 stars)



Share of 1-2 star reviews by delivery delay



Late vs on-time orders: the gap is large and significant

Metric	On time	Late	Gap	Bootstrap 95% CI	p-value
Average review (stars)	4.28	2.22	2.06	2.02 to 2.10	< 0.001
Share of 1-2 star reviews	9.4%	63.7%	54.3%	53.1% to 55.5%	< 0.001

Repurchase within 180 days, logistic regression (odds ratio < 1 = less repurchase; a CI that includes 1 = not significant)

Model	Variable	Odds ratio	95% CI	p-value
A: late + controls	Late first order	0,84	0.62 to 1.13	0.240
A: late + controls	Lives in São Paulo state	1,19	1.05 to 1.36	0.007
A: late + controls	Order value (log)	0,99	0.91 to 1.07	0.726
B: A + review score	Late first order	0,97	0.70 to 1.33	0.827
B: A + review score	Lives in São Paulo state	1,18	1.04 to 1.35	0.010
B: A + review score	Order value (log)	0,99	0.92 to 1.07	0.827
B: A + review score	Review score (+1 star)	1,06	1.00 to 1.12	0.036

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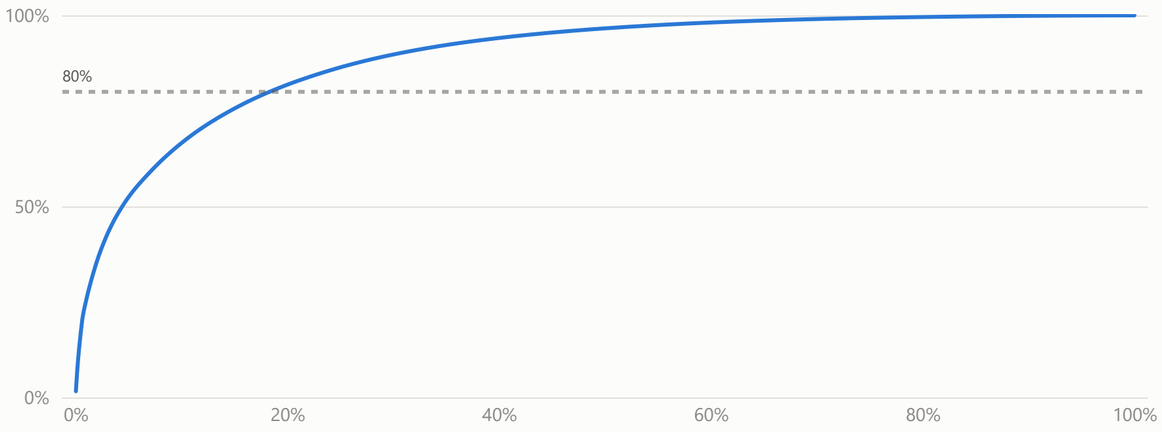
18% of sellers make 80% of GMV; 32 big sellers with poor ratings are the ones to fix first

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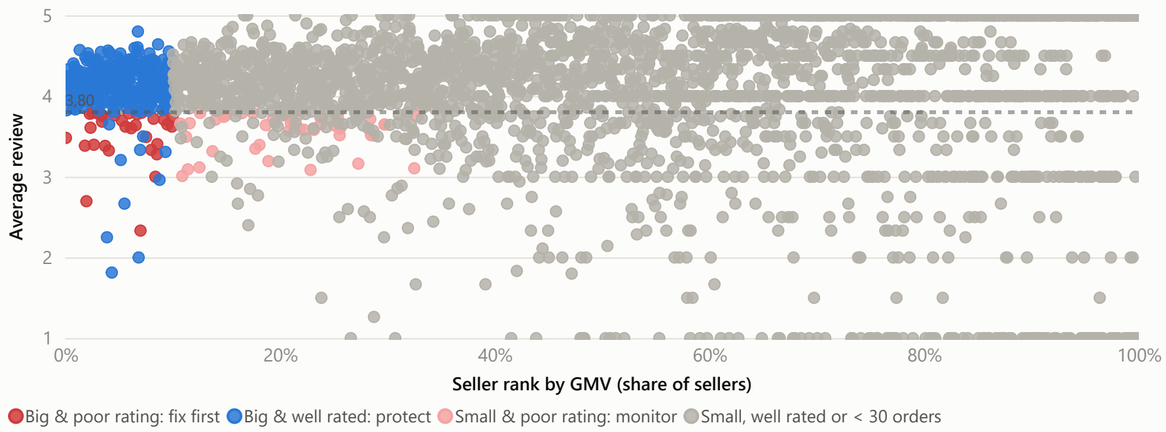
The top 10% of sellers carry 67% of GMV. The 32 big sellers rated below 3.8 (with at least 30 orders) hold 6.3% of GMV and deliver late 10% of the time, vs 7% overall.

Sellers with sales	Share of sellers making 80% of GMV	Big sellers rated below 3.8	Their share of GMV
2 878	18%	32	6,3%

Cumulative share of GMV by seller rank (Pareto curve, dashed = 80%)



Each seller: rank by GMV (left = biggest, first 10% = 'big') vs average review (dashed = 3.8)



Fix first: the 32 big sellers (top 10% GMV) rated below 3.8 with 30+ orders, largest first

Seller	State	Orders	GMV (BRL)	Late rate	Avg review	1-2 star share
7c67e1448b00f6e969d365cea6b010ab	SP	973	237 792	9,1%	3,48	25,9%
712e6ed8aa4aa1fa65dab41fed5737e4	SC	79	46 704	20,8%	3,39	34,6%
2eb70248d66e0e3ef83659f71b244378	SP	202	45 860	11,2%	2,70	50,0%
855668e0971d4dfd7bef1b6a4133b41b	SP	307	39 460	12,8%	3,78	22,5%
1835b56ce799e6a4dc4eddc053f04066	SP	391	39 250	10,9%	3,61	25,3%
70a12e78e608ac31179aea7f8422044b	SP	313	36 239	9,4%	3,79	20,3%
88460e8ebdecbfecb5f9601833981930	PR	233	34 856	17,7%	3,40	31,0%

Seller groups: size and quality (big = top 10% by GMV; poor = average review below 3.8, 30+ orders)

Group	Sellers	GMV share	Avg seller review	Avg seller late rate
Big & poor rating: fix first	32	6,3%	3,52	10,4%
Big & well rated: protect	255	60,2%	4,12	6,8%
Small & poor rating: monitor	45	1,8%	3,57	12,5%
Small, well rated or < 30 orders	2 546	31,7%	4,05	6,9%

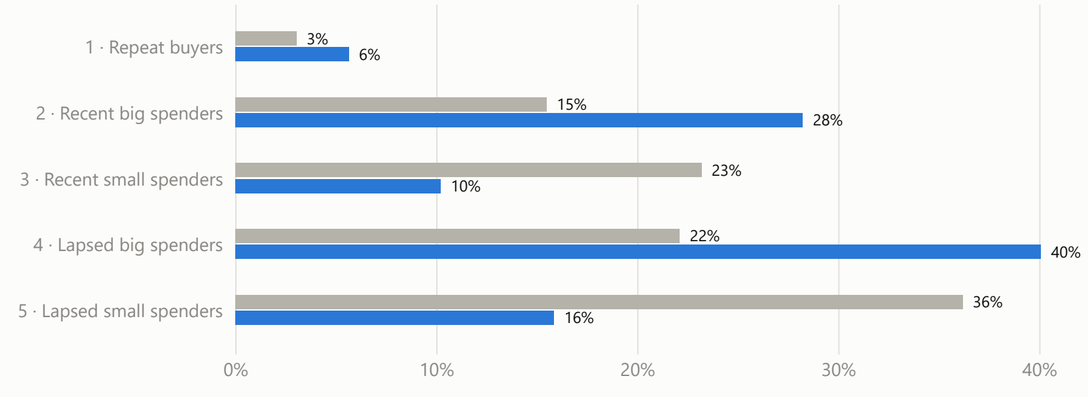
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Lapsed big spenders are 22% of customers but 40% of GMV: the first win-back target

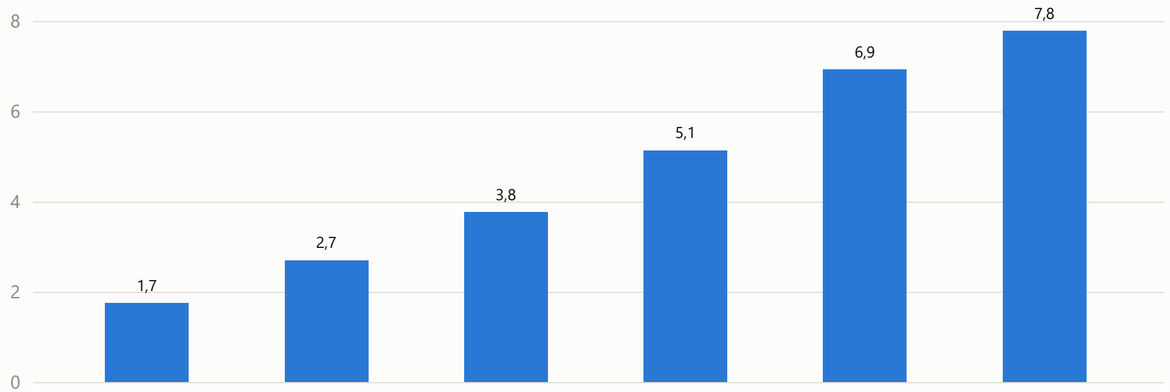
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RFM segments rest on Recency and Monetary quintiles: with 97% one-time buyers, Frequency only separates the 3% who came back. Card installments grow with the basket (Spearman 0.46): Brazilian shoppers spread big purchases over months.

RFM segments: share of customers (grey) vs share of GMV (blue)



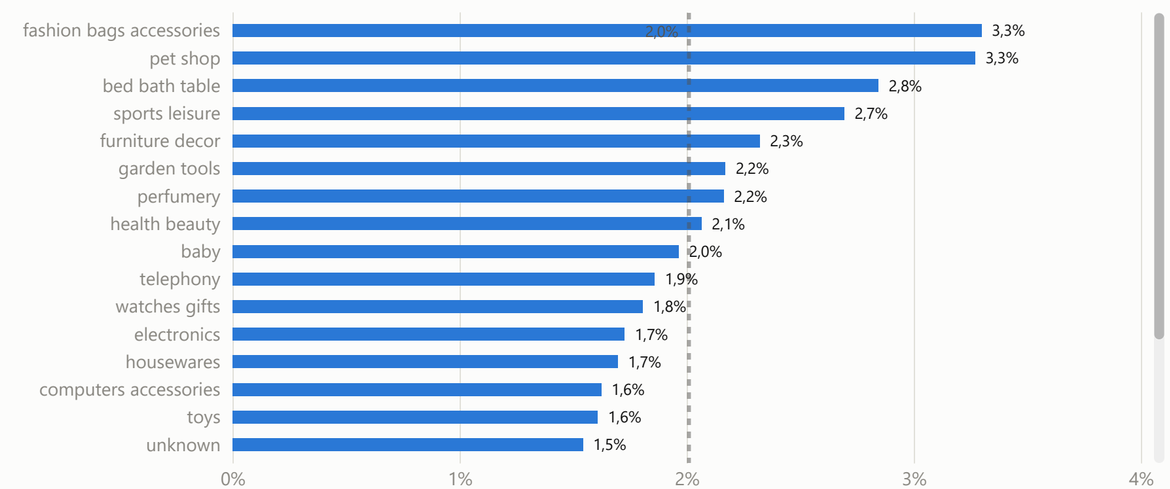
Average card installments by order value, BRL (overall average 3.5)



Segments at 1 Aug 2018 and their top-3 categories by GMV

Segment	Customers	Avg spend (BRL)	Days since last order	Top categories
1 · Repeat buyers	2 702	307	208	bed bath table, sports leisure, furniture decor
2 · Recent big spenders	13 744	303	85	watches gifts, health beauty, computers accessories
3 · Recent small spenders	20 585	73	86	health beauty, bed bath table, sports leisure
4 · Lapsed big spenders	19 624	301	319	watches gifts, health beauty, computers accessories
5 · Lapsed small spenders	32 119	73	321	bed bath table, sports leisure, health beauty

Repurchase within 180 days by first-order category (dashed = 2.0% average)

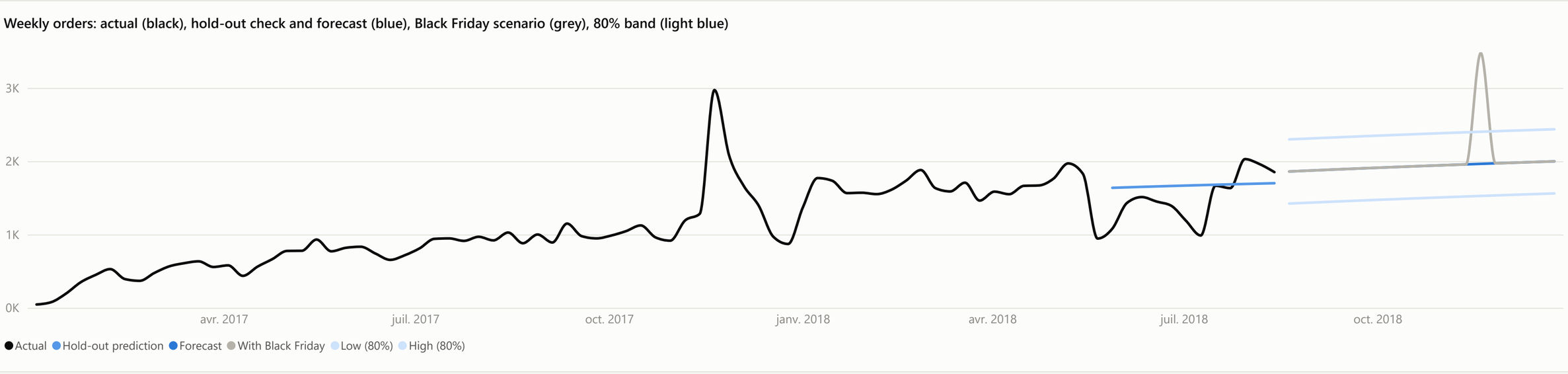


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Q4 2018: about 25,400 orders expected (27,000 with a Black Friday spike), +46% on Q4 2017

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Damped Holt trend, checked on 12 hidden weeks: 22% average weekly error, no better than a naive forecast (21%), so plan on the 80% band (19,800-31,100 orders) rather than the point. Black Friday 2017 was x1.8 a normal week.



Hold-out error by model (last 12 weeks)

Model	MAPE	MAE (orders)
Naive (last week)	21,3%	280
Holt, damped trend	21,8%	281
Mean of last 8 weeks	24,2%	312
Linear trend	37,6%	484

RECOMMENDATION 1

Cut lateness, starting with the big, poorly rated sellers

Late orders are 7% of orders but 33% of all 1-2 star reviews. 32 top-10% sellers rate below 3.8 (late rate 10%).

Size Halving lateness = ~1,627 fewer 1-2 star reviews (14% of them)

RECOMMENDATION 2

Win back one-time buyers with a second-order offer

97% of customers ordered once; only 2.0% reorder within 180 days.

Size +1 pt of repurchase = ~707 extra orders and ~BRL 113k GMV per year (70,688 new customers/yr)

RECOMMENDATION 3

Acquire through the categories that bring customers back

Best first-order categories: fashion bags accessories, pet shop, bed bath table (2.8%-3.3% vs 2.0% average).

Size Up to x1.6 the average repurchase rate per customer acquired

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